Summary

- According to Arrhenius, an acid is a substance that dissociates to give hydrogen ions in water.
- According to Lowry and Bronsted concept, an acid is defined as a substance that has a tendency to donate a proton to another substance and base is a substance that has a tendency to accept a proton from other substance.
- According to Gilbert . N. Lewis , an acid is a species that accepts an electron pair while base is a species that donates an electron pair.
- ionic product (ionic product constant) of water (K_w)=$[H_3O^+][OH^-]$
- pH of a solution is defined as the negative logarithm of base 10 of the molar concentration of the hydronium ions present in the solution.

$$pH = -\log_{10}[H_3O^+]$$

- when dilution increases, the degree of dissociation of weak electrolyte also increases. This statement is known as Ostwald's dilution Law.
- When a salt of a weak acid is added to the acid itself, the dissociation of the weak acid is suppressed further this is known as common ion effect
- Buffer is a solution which consists of a mixture of a weak acid and its conjugate base (or) a weak base and its conjugate acid.
- Buffer capacity and buffer index is defined as the number of gram equivalents of acid or base added to 1 litre of the buffer solution to change its pH by unity.

$$\beta = \frac{dB}{d(pH)}$$

- Henderson – Hasselbalch equation

For Acid buffer

$$\Rightarrow pH = pK_a + \log\frac{[salt]}{[acid]}$$

For Basic buffer

$$\Rightarrow pOH = pK_b + \log\frac{[salt]}{[base]}$$

- Hydrolysis of Salt of strong base and weak acid

$$K_h.K_a = K_w$$

$$pH = 7 + \tfrac{1}{2}pK_a + \tfrac{1}{2}\log C.$$

- Hydrolysis of salt of strong acid and weak base

$$K_h.K_b = K_w$$

$$pH = 7 - \tfrac{1}{2}pK_b - \tfrac{1}{2}\log C.$$

- Hydrolysis of Salt of weak acid and weak base

$$K_a.K_b.K_h = K_w$$

$$pH = 7 + \tfrac{1}{2}pK_a - \tfrac{1}{2}pK_b.$$

- The solubility product of a compound is defined as the product of the molar concentration of the constituent ions, each raised to the power of its stoichiometric co – efficient in a balanced equilibrium equation.

XII U8-Ionic equilibrium.indd 27 16-12-2022 19:34:26